Geometry-Aware JEPA for EEG

A controlled frozen-transfer study on TUAB

Florent Tariolle   Clément Genninasca   Yoann Frayce   Hippolyte du Pac de Marsoulies

Hack the World(s) – EEG Track · Team **HelloWorlds**

Contribution

Our contribution, in one slide

1. **A strong frozen baseline.** SIGReg-ambient JEPA matches the LeJEPA/Laya recipe's value on our (pipeline-helped) setup – not a controlled head-to-head.
2. **Nothing beats it.** Ambient/tangent SIGReg & PEIRA, Graph-JEPA, Fourier-JEPA – none beats ambient SIGReg in balanced accuracy (gaps ≤ 0.013).
3. **Latent geometry is the contribution.** Ours separates normal/abnormal $\sim 2\times$ better than the random-init encoder*, visualised with **de Surrel's AIRM Riemannian t-SNE** (the right metric for SPD/EEG covariances). Tangent sharpens the *geometry* (best cov-latent silhouette), not the task.
4. **Rigor over spin.** The limits we report – *decodable* $\neq$ *clustered*, sustained vs transient, AIRM edge within noise. An honest negative beats a fragile positive.

* *Random-init encoder* = identical architecture, weights never trained, encoder frozen; only the linear probe is fit – it isolates the SSL contribution.

Motivation

The question we actually asked

- **EEG foundation models** are everywhere – but most report *fine-tuned* numbers. What survives when you **freeze** the encoder?
- **Self-supervised JEPA** avoids collapse with an *anti-collapse regulariser*. The EEG covariance lives on a curved **SPD manifold**, not in flat Euclidean space.
- **Our pre-registered hypothesis**: making anti-collapse *geometry-aware* (act in the SPD **tangent**) and *distribution-free* (**PEIRA**) beats a plain ambient baseline.
- **What we ship**: a controlled study with an *honest* answer – a strong frozen probe, a clean negative result with a geometric mechanism, and label/data-efficiency curves.

Data

Data: TUH Abnormal EEG (TUAB)

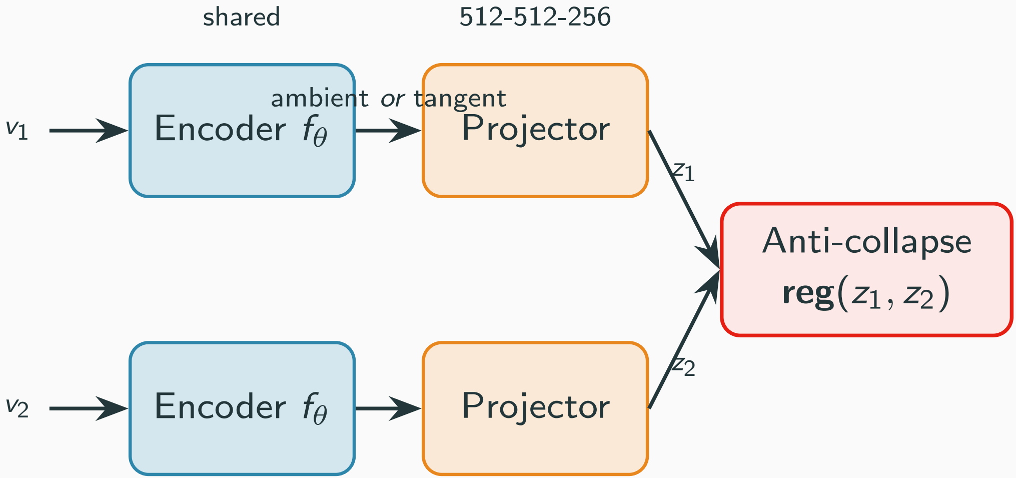
- **Task**: binary *normal* vs *abnormal* EEG, recording level.
- **Signal**: 19 channels @ 200 Hz; we sample 10 s windows (2,000 samples), per-channel z-scored.
- **Standard split**: 2,717 train / 276 eval recordings, **patient-disjoint** (2,076 vs 253 patients, overlap = 0 – verified).
- **Pretraining**: SSL reads only *unlabeled* train recordings; labels enter at the probe alone.

Why we don't chase accuracy

TUAB labels come from clinical reports and are known to be noisy; we are not aware of a published label-noise / Bayes-ceiling estimate. **Sub-1-point gaps without multi-seed CIs are not meaningful** – so we report *balanced accuracy on the full split*, 3 seeds, with error bars.

Architecture

Two view and 1D convolutional encoder



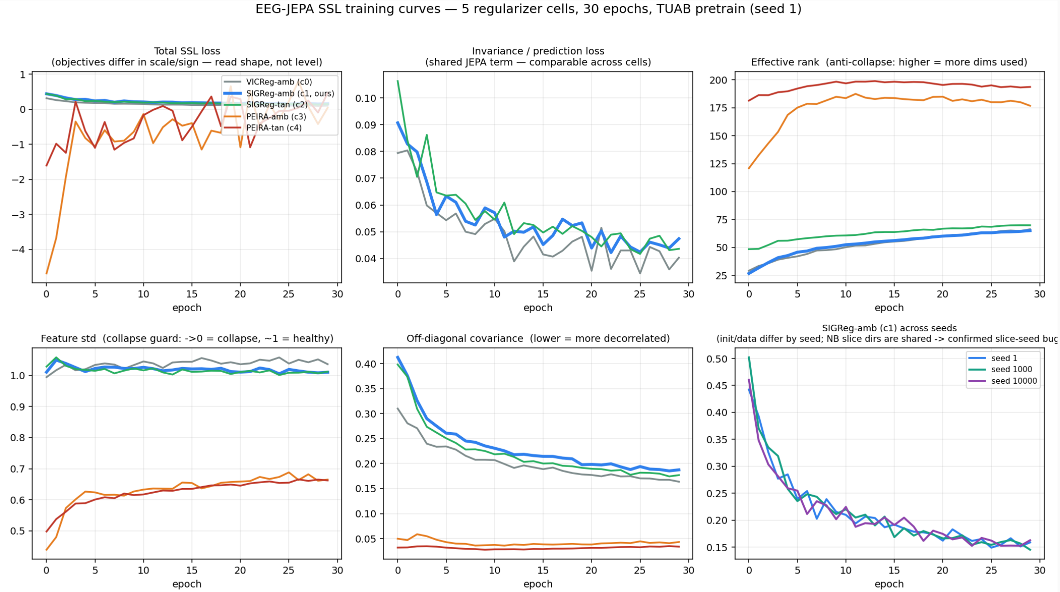
What we tried

regulariser \ **space**	**ambient** (Euclidean)	**tangent** (SPD)
VICReg (variance/cov)	`C0` reference	—
SIGReg (Gaussianity slices)	`C1` *Laya-like baseline*	`C2` geometry-aware
PEIRA (distribution-free)	`C3`	`C4` *ex-hypothesis*

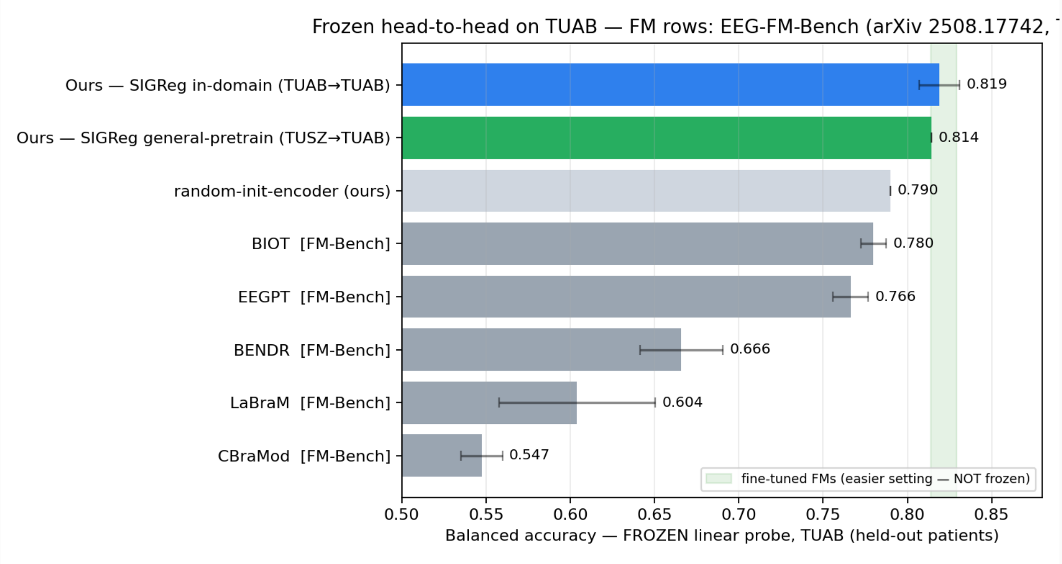
- **SIGReg** (LeJEPA/BCS): tests random 1-D slices for Gaussianity – assumes an *isotropic* target.
- **PEIRA**: distribution-free anti-collapse – no target distribution to mis-specify.
- **Hypothesis**: the geometry-aware / distribution-free cells (`C2`, `C4`) beat the ambient baseline `C1`.
- **GraphJEPA**
- **FourierJEPA**

Results

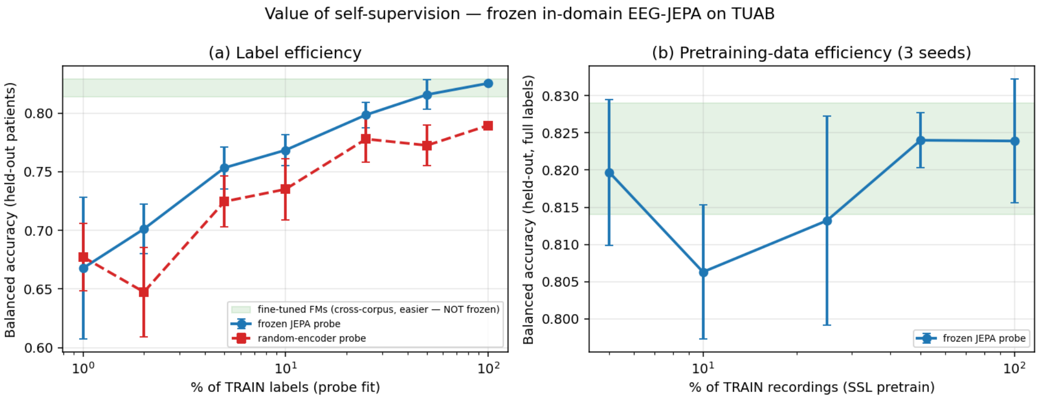
SSL training curves (all cells healthy)



Frozen & in-domain: we hold where frozen FMs fall

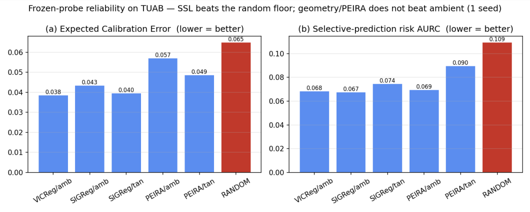


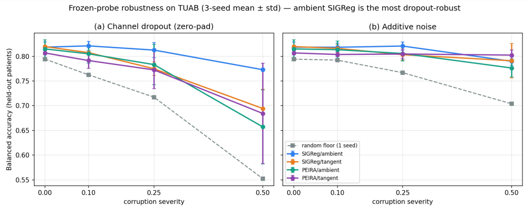
The value of self-supervision (label & data efficiency)



(a) The money plot: *JEPA @ 25% labels* $\approx$ *random-init encoder @ 100%* – $\sim 4\times$ label efficiency, $\sim +0.04$ BalAcc. **(b)** Probe is stable down to small pretraining-data fractions.

Beyond accuracy: calibration & robustness (3 axes, not 1)





- **Calibration**: every SSL encoder beats the random-init-encoder floor (ECE, selective prediction) – geometry/PEIRA does *not* beat ambient.
- **Robustness (3-seed)**: ambient SIGReg is the *most* channel-dropout-robust; the SPD-tangent objective *degrades* dropout-robustness; no geometry win under noise (the single-seed hint did not replicate).
- **The null holds on all three axes** – accuracy, calibration, robustness. Geometry buys none of them here.

Visualization

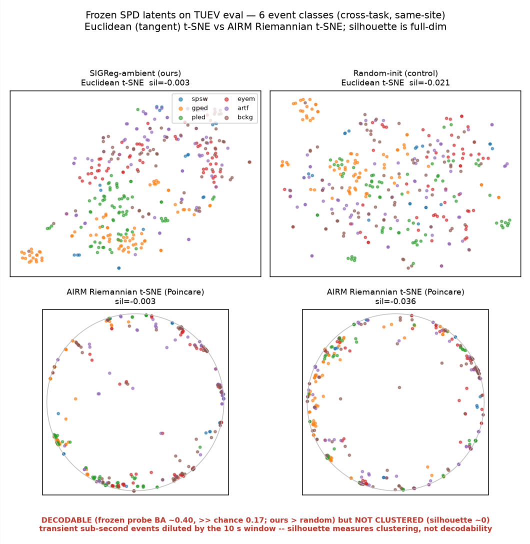
Manifold-aware latent geometry (AIRM vs Euclidean)

Frozen SPD covariance latent, two embeddings: Euclidean (tangent) t-SNE vs **AIRM Riemannian**

Manifold viz across TUH tasks – sustained vs transient

TUSZ – 5 classes; SIGReg-tangent separates best

TUEV – 6 event types (decodable, not clustered)



task (#cls)	sil_{Euc}	sil_{AIRM}

- Clusters **sustained** phenomena (abnormality, seizures), not

Caveats we disclose to the jury

- **Report 0.819, not 0.833.** The single-seed 0.833 (SIGReg-ambient, seed 1) was the *top* of that cell's range; the 3-seed mean is 0.819.
- **SIGReg slice-seed bug (found, fixed).** `BCS` seeded slice directions per-step, so the 3 SIGReg seeds shared one slice schedule – error bars *understate* variance. Now fixed (run-seeded RNG); the reported numbers predate the fix, not re-run.
- **PEIRA tangent (0.807) is not evidence against wrapped-Gaussian.** PEIRA is distribution-free $\Rightarrow$ immune to the isotropic-target mis-specification $\Rightarrow$ it never tests that hypothesis. 0.807 vs 0.815 is within noise.
- **Bootstrap CI (illustrative, not yet run).** A bootstrap over the 276 eval recordings would plausibly give a $\pm$0.02–0.03 CI – wider than the max inter-cell gap (0.013) – supporting "benchmark-resolution-limited". The resampling is not yet committed.
- **Preprocessing differs** from the published FM rows – the head-to-head is indicative, not a controlled re-run of their pipelines.

Limitations & Future Work

Limitations & Future Work

- **Single dataset.** Specialization $\neq$ generality. The honest generality test is frozen cross-dataset transfer (TUEV/Sleep-EDF) – started, but left as future work rather than over-claimed.
- **Underpowered null.** $n{=}3$ seeds; the inter-cell gaps (≤ 0.013) are below a plausible eval bootstrap CI ($\pm$0.02–0.03, not yet computed). The honest statement is "*benchmark-resolution-limited*", not "geometry is dead".
- **The principled fix.** A wrapped-Gaussian SIGReg on the *channel* covariance, AIRM tangent at the running Fréchet mean – $\sim$40 LOC + one re-pretrain, with real `eigh`-backward numerical risk.
- **Where geometry should pay off.** Robustness, calibration, and transfer under montage/threshold shift – *not* a balanced-accuracy race.

Conclusion

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- A **controlled frozen-transfer study**, not a leaderboard entry.
- **A strong frozen probe** (0.819 BA, $\sim$0.89 AUROC) that holds where published FMs collapse when frozen – with the random-init-encoder floor honestly disclosed.
- **A clean, pre-registered negative result**: *where* anti-collapse acts (ambient vs SPD-tangent) and *which* regulariser do not change frozen-probe accuracy on TUAB – with a geometric reason why.
- **The null is multi-axis**: geometry/PEIRA help neither accuracy, nor calibration, nor robustness (3 seeds) – and it holds even *general*-pretrained (TUSZ $\rightarrow$ frozen TUAB, 0.814).
- **Honesty as the headline**: an honest negative beats a fragile positive.

Built on a trimmed `eb_jepa` vendor. Team **Hello Worlds**.

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